

Dataset: Joint UAV Routing and Task Offloading Optimization Problem Formulation for Smart Manufacturing Systems

Abstract—This document presents a mathematical optimization problem formulation for a smart manufacturing system that integrates UAV routing and computational task offloading. In this scenario, a fleet of unmanned aerial vehicles (UAVs) is deployed to collect finished products from geographically distributed manufacturing stations while simultaneously providing mobile edge computing (MEC) services to industrial sensor devices (ISDs). The formulation captures the joint decision-making process across multiple dimensions, including UAV routing, product collection, task execution mode selection (local, UAV-based, or cloud-assisted), and computing and communication resource allocation over a finite time horizon. The objective function is designed to maximize the total collection value and the number of tasks completed within their deadlines, while minimizing deadline violations, task completion delay (flow time), and overall resource consumption. The optimization problem is subject to a comprehensive set of constraints, including UAV routing feasibility, payload and travel limits, mission time constraints, service availability, communication and computing capacity limits, task completion requirements, and UAV energy budgets. This document provides a complete optimization problem formulation, including the objective function, constraints, decision variables, parameters, and notation. It is intended to serve as a reference for retrieving mathematical models related to UAV routing, task offloading, and integrated optimization in smart manufacturing and MEC-enabled systems.

I. PROBLEM STATEMENT

A. System Description

A typical manufacturing sequence includes: **material delivery** $\rightarrow$ **manufacturing completion** $\rightarrow$ **product collection**. Accordingly, the logistics process consists of two phases: **phase 1** — **material delivery** and **phase 2** — **product collection**. In this paper, we focus on phase 2. Specifically, a fleet of U homogeneous unmanned aerial vehicles (UAVs) is dispatched from a central depot equipped with terrestrial mobile edge computing (MEC) servers, hereafter referred to as the cloud, to collect finished products from M geographically distributed manufacturing stations while providing MEC services to industrial sensor devices (ISDs) at these stations. All UAVs depart from and return to the depot, and fly at a fixed altitude with constant speed. Each station $m \in S$ is located at (X_m, Y_m) , has a collection reward v_m , product weight w_m , and hosts a set of ISDs K_m . All finished products are assumed ready for collection at the mission start. Over a discretized time horizon, ISDs generate computational tasks stochastically according to industrial requirements.

B. UAV Constraints and Operational Mode

Each UAV is constrained by battery energy, maximum travel distance, payload capacity, and onboard computing and communication resources. Its total energy consumption includes flight, hovering, computation, and communication. The mission follows a depot-based mode, i.e., **depot** $\rightarrow$ **manufacturing stations** $\rightarrow$ **depot**, and each UAV may visit multiple stations subject to its payload and travel limits. During station service, a UAV simultaneously collects finished products and provides MEC services to local ISDs. Hence, routing decisions jointly determine collection performance and MEC availability, since UAV service is feasible only when the UAV is present and within communication range. A collection task is completed when the products are delivered to the depot, whereas a computational task is successful if it is completed before its deadline via local processing, UAV-based MEC, or cloud processing.

C. Three-Layer MEC Architecture

The computational task execution follows a three-layer cloud-edge-device architecture consisting of the device layer, the aerial MEC layer, and the cloud layer. In the device layer, ISDs stochastically generate computational tasks and may process them locally. In the aerial MEC layer, a UAV can process tasks from local ISDs when serving a station. In the cloud layer, tasks may be offloaded to the cloud through a UAV-assisted path, where the ISD uploads task data to a serving UAV and the UAV forwards it over the backhaul. Each task is executed by exactly one processor, i.e., the ISD, the UAV, or the cloud.

D. Optimization Objectives and Constraints

We study a finite-horizon joint optimization problem that couples UAV product collection with MEC task scheduling. The goal is to jointly determine UAV routes, station service, task execution modes, and resource allocation to collect high-value products while completing ISD tasks in a timely manner. The optimization is subject to: (i) depot-based routing and collection assignment, with each station served by at most one UAV; (ii) UAV payload, travel-distance, mission-time, and battery-energy limits; (iii) computing, communication, and deadline constraints for task execution; and (iv) communication link capacity and service availability constraints for UAV-assisted offloading.

II. MATHEMATICAL FORMULATION

In this section, we define the notation, objective function, and constraints associated with UAV routing and task offloading.

A. Definition of Notation

Let $L = \{0\} \cup S$ be the routing node set, where node 0 denotes the depot, and let $\mathcal{T} = \{0, \dots, T-1\}$ be the slot set with slot length Δ . Each realized task is indexed by $(k, \tau) \in \mathcal{I}$, meaning that ISD k generates a task at slot τ . For routing, $x_{u,i,j} \in \{0, 1\}$ indicates whether UAV u flies directly from node i to node j , $y_{u,m} \in \{0, 1\}$ indicates whether station m is assigned to UAV u , and $c_m \in \{0, 1\}$ indicates whether the product at station m is collected. For timing, $a_{u,m}$ and $d_{u,m}$ denote the arrival and departure times of UAV u at station m , while s_u and r_u denote its departure time from and return time to the depot. For task service, $\eta_{u,m,t} \in \{0, 1\}$ indicates whether UAV u serves station m in slot t . For execution-mode selection, $\ell_{k,\tau}$, $p_{k,u,\tau}$, and $g_{k,u,\tau}$ denote local execution, UAV execution, and cloud execution via UAV u , respectively, and we define $h_{k,u,\tau} \triangleq p_{k,u,\tau} + g_{k,u,\tau}$, which indicates whether task (k, τ) uses UAV u to receive its input data, either for execution at UAV u or for forwarding to the cloud. The relevant decision variables, parameters, and other symbols are summarized in Tables I and II.

B. Objective Function

Let $T_{k,\tau}$ be the completion time of task (k, τ) , and let $z_{k,\tau} \in \{0, 1\}$ indicate whether the task is completed before its deadline. Also, let R_t^{cmp} and R_t^{com} denote the normalized computing and communication occupation in slot t , respectively. The objective is defined in Eq. (1).

$$\begin{aligned}
 \max \quad & \underbrace{\omega_{\text{col}} \sum_{m \in S} v_m c_m}_{\text{collection value}} + \underbrace{\omega_{\text{cmp}} \sum_{(k,\tau) \in \mathcal{I}} z_{k,\tau}}_{\text{on-time completion return}} \\
 & - \underbrace{\omega_{\text{miss}} \sum_{(k,\tau) \in \mathcal{I}} (1 - z_{k,\tau})}_{\text{deadline violation cost}} - \underbrace{\omega_{\text{flow}} \sum_{(k,\tau) \in \mathcal{I}} (T_{k,\tau} - \tau \Delta)}_{\text{flow-time cost}} \\
 & - \underbrace{\omega_{\text{res}} \sum_{t \in \mathcal{T}} (R_t^{\text{cmp}} + R_t^{\text{com}})}_{\text{resource-occupation cost}}. \quad (1)
 \end{aligned}$$

The objective is to balance the logistics and computing performance by promoting product collection and deadline-compliant task completion, while reducing deadline violations, task completion delay, and overall computing and communication resource occupation.

C. Constraints from UAV Routing

The UAV-routing constraints are organized into three parts, namely collection assignment and route consistency, payload and distance budgets, and mission timing feasibility.

TABLE I
SYMBOLS: DECISION VARIABLES AND PARAMETERS

Symbols	Description
Decision Variables	
$a_{u,m}, d_{u,m}$	Arrival / departure time of UAV u at station m .
$b_{k,u,\tau,t}^{\text{bh}}$	UAV- $u \rightarrow$ cloud rate allocated (bits/s).
$b_{k,u,\tau,t}^{\text{ul}}$	ISD $\rightarrow$ UAV- u rate allocated (bits/s).
c_m	Binary: station m collected.
$\delta_{k,\tau,t}$	Binary: task (k, τ) completes in slot t .
$f_{k,\tau,t}^{\text{loc}}$	Local compute allocated (work-unit/s).
$f_{k,u,\tau,t}^{\text{cld}}$	Cloud compute via UAV u allocated (work-unit/s).
$f_{k,u,\tau,t}^{\text{uav}}$	UAV- u compute allocated (work-unit/s).
$g_{k,u,\tau}$	Binary: task (k, τ) executed in cloud via UAV u .
$h_{k,u,\tau}$	Binary: UAV u is used as first hop for task (k, τ) .
$\ell_{k,\tau}$	Binary: task (k, τ) processed locally.
$p_{k,u,\tau}$	Binary: task (k, τ) processed on UAV u .
s_u, r_u	Depot departure / return time of UAV u .
$T_{k,\tau}$	Task completion time.
$x_{u,i,j}$	Binary: UAV u travels $i \rightarrow j$.
$y_{u,m}$	Binary: station m assigned to UAV u .
$z_{k,\tau}$	Binary: task (k, τ) meets deadline.
$\eta_{u,m,t}$	Binary: UAV u serves station m in slot t .
Parameters	
$B_{k,\tau}$	Task input size (bits).
$B_u^{\text{ul}}, B_u^{\text{bh}}$	UAV u uplink / backhaul limits (bits/s).
$d_{i,j}$	Distance between nodes i and j .
D_k	Task deadline duration for ISD k tasks.
D_u	UAV distance budget.
E_u^{max}	UAV u energy budget.
F^{cld}	Cloud compute capacity (work-unit/s).
F_k^{loc}	ISD k local compute capacity (work-unit/s).
F_u^{uav}	UAV u compute capacity (work-unit/s).
Q_u	UAV payload capacity.
T_{mission}	Mission horizon.
$v_{\text{fly}}, v_{\text{hov}}$	UAV flight speed.
v_m, w_m	Product value / weight at station m .
$W_{k,\tau}$	Task workload (work-units).
$\alpha_{\text{bh}}, \alpha_{\text{cmp}}, \alpha_{\text{fly}}, \alpha_{\text{hov}}, \alpha_{\text{ul}}$	Energy coefficients for UAV-to-cloud communication, onboard computing, flying and hovering, and ISD-to-UAV communication.
Δ	Time slot duration.
$\gamma_{k,u,t}^{\text{ul}}, \gamma_{u,t}^{\text{bh}}$	Effective-rate factors (uplink / backhaul).
$\omega_{\text{col}}, \omega_{\text{cmp}}$	Weights for collection value and task completion.
$\omega_{\text{miss}}, \omega_{\text{flow}}$	Weights for miss penalty and flow-time.
ω_{res}	Weight for resource-occupation cost.
τ_m	Minimum service time at station m .

a) Collection assignment and route consistency:

$$\sum_{u \in U} y_{u,m} \leq 1, \quad c_m \leq \sum_{u \in U} y_{u,m}, \quad \forall m \in S, \quad (2)$$

$$\sum_{i \in L} x_{u,i,m} = \sum_{j \in L} x_{u,m,j} = y_{u,m}, \quad \forall u \in U, \forall m \in S, \quad (3)$$

$$\sum_{m \in S} x_{u,0,m} = \sum_{m \in S} x_{u,m,0}, \quad \forall u \in U. \quad (4)$$

Eq. (2) limits each station to at most one UAV and allows a station to be counted as collected only if it is assigned. Eq. (3) ensures that station m appears on the route of UAV u exactly when it is assigned to UAV u , in which case it is entered and left exactly once by UAV u . Eq. (4) ensures that each UAV

TABLE II
SYMBOLS: SUPERSCRIPTS AND OTHERS

Symbols	Description
Superscripts	
cmp, com	Computing-related / communication-related.
fly, hov	Flight / hovering energy tag.
loc, uav, cld	Local / UAV / cloud mode or capacity type.
ul, bh	Uplink (ISD→UAV) / backhaul (UAV→cloud).
max	Budget upper bound.
Others	
S, U, K	Sets of stations / UAVs / ISDs.
L	Set of routing nodes, $L = \{0\} \cup S$, 0 denotes depot.
$\mathcal{I}$	Realized-task set indexed by (k, τ) .
$\mathcal{T}$	Time slot set; $t \in \mathcal{T}$.
T	Total number of time slots in $\mathcal{T} = \{0, \dots, T-1\}$.
$\chi_{k,\tau}(t)$	Cumulative completion flag by slot t .
$m(k)$	Station hosting ISD k .
R_t^{cmp}	Normalized computing occupation in slot t .
R_t^{com}	Normalized communication occupation in slot t .
E_u^{fly}	UAV flight-hover energy.
$E_u^{\text{cmp}}, E_u^{\text{com}}$	UAV onboard-computing / communication energy.
m, u, k, i, j	Indices: station / UAV / ISD / node i / node j .
τ, t	Indices: task-generation slot / slot.

departs from and returns to the depot the same number of times.

b) *Payload and distance budgets:*

$$\sum_{m \in S} w_m y_{u,m} \leq Q_u, \quad \sum_{i \in L} \sum_{j \in L} d_{i,j} x_{u,i,j} \leq D_u, \quad \forall u \in U. \quad (5)$$

Eq. (5) limits the total assigned payload weight and the total travel distance of each UAV.

c) *Mission timing:*

$$s_u = 0, \quad r_u \leq T_{\text{mission}}, \quad \forall u \in U, \quad (6)$$

$$x_{u,0,m} = 1 \Rightarrow a_{u,m} \geq s_u + \frac{d_{0,m}}{v_{\text{fly}}}, \quad \forall u \in U, \forall m \in S, \quad (7)$$

$$x_{u,i,j} = 1 \Rightarrow a_{u,j} \geq d_{u,i} + \frac{d_{i,j}}{v_{\text{fly}}}, \quad \forall u \in U, \forall i, j \in S, \quad (8)$$

$$x_{u,m,0} = 1 \Rightarrow r_u \geq d_{u,m} + \frac{d_{m,0}}{v_{\text{fly}}}, \quad \forall u \in U, \forall m \in S, \quad (9)$$

$$d_{u,m} - a_{u,m} \geq \tau_m y_{u,m}, \quad \forall u \in U, \forall m \in S. \quad (10)$$

Eq. (6) states that each UAV starts at time 0 and must return within the mission horizon. Eqs. (7) to (9) ensure that the route timing is consistent with the corresponding travel times. In Eq. (8), the case $i = j$ is excluded, since no travel is needed from a station to itself. Eq. (10) requires a minimum service duration at each station assigned to UAV u .

D. *Constraints from Task Offloading*

The constraints associated with task offloading include execution-mode selection, completion-time encoding, computing and communication resource limits, service-window indicators, service-window feasibility for UAV-involved processing, deadline satisfaction, and UAV energy budgeting.

a) *Execution-mode selection:*

$$\ell_{k,\tau} + \sum_{u \in U} p_{k,u,\tau} + \sum_{u \in U} g_{k,u,\tau} = 1, \quad \forall (k, \tau) \in \mathcal{I}. \quad (11)$$

Eq. (11) requires every task to choose exactly one execution mode: local execution, UAV execution, or cloud execution via a UAV relay.

b) *Completion-time encoding:*

$$z_{k,\tau} = \sum_{t=\tau}^{T-1} \delta_{k,\tau,t}, \quad \forall (k, \tau) \in \mathcal{I}, \quad (12)$$

$$T_{k,\tau} = \sum_{t=\tau}^{T-1} (t+1) \Delta \delta_{k,\tau,t} + T_{\text{mission}} (1 - z_{k,\tau}). \quad (13)$$

Eqs. (12) and (13) encode the completion time $T_{k,\tau}$ and the completion-status indicator $z_{k,\tau}$ via the slot-selection binary $\delta_{k,\tau,t} \in \{0, 1\}$.

c) *Computing and communication resource constraints:*

The computing and communication resources are subject to three types of constraints, namely per-slot capacity limits, mode consistency, and cumulative service sufficiency.

(i) *Per-slot capacity limits*

$$\begin{cases} \sum_{\tau': (k, \tau') \in \mathcal{I}, \tau' \leq t} f_{k,\tau',t}^{\text{loc}} \leq F_k^{\text{loc}}, & \forall k \in K, \forall t \in \mathcal{T}, \\ \sum_{(k, \tau) \in \mathcal{I}: \tau \leq t} f_{k,u,\tau,t}^{\text{uav}} \leq F_u^{\text{uav}}, & \forall u \in U, \forall t \in \mathcal{T}, \\ \sum_{u \in U} \sum_{(k, \tau) \in \mathcal{I}: \tau \leq t} f_{k,u,\tau,t}^{\text{cld}} \leq F^{\text{cld}}, & \forall t \in \mathcal{T}. \end{cases} \quad (14)$$

$$\begin{cases} \sum_{(k, \tau) \in \mathcal{I}: \tau \leq t} b_{k,u,\tau,t}^{\text{ul}} \leq B_u^{\text{ul}}, & \forall u \in U, \forall t \in \mathcal{T}, \\ \sum_{(k, \tau) \in \mathcal{I}: \tau \leq t} b_{k,u,\tau,t}^{\text{bh}} \leq B_u^{\text{bh}}, & \forall u \in U, \forall t \in \mathcal{T}. \end{cases} \quad (15)$$

Eqs. (14) and (15) bound the aggregate computing and communication resource allocations in each slot by the available local, UAV, cloud, uplink, and backhaul capacities.

(ii) *Mode Consistency*

$$\begin{cases} 0 \leq f_{k,\tau,t}^{\text{loc}} \leq F_k^{\text{loc}} \ell_{k,\tau}, & \forall (k, \tau) \in \mathcal{I}, \forall t \geq \tau, \\ 0 \leq f_{k,u,\tau,t}^{\text{uav}} \leq F_u^{\text{uav}} p_{k,u,\tau}, & \forall (k, \tau) \in \mathcal{I}, \forall u \in U, \forall t \geq \tau, \\ 0 \leq f_{k,u,\tau,t}^{\text{cld}} \leq F^{\text{cld}} g_{k,u,\tau}, & \forall (k, \tau) \in \mathcal{I}, \forall u \in U, \forall t \geq \tau, \end{cases} \quad (16)$$

$$\begin{cases} 0 \leq b_{k,u,\tau,t}^{\text{ul}} \leq B_u^{\text{ul}} h_{k,u,\tau}, & \forall (k, \tau) \in \mathcal{I}, \forall u \in U, \forall t \geq \tau, \\ 0 \leq b_{k,u,\tau,t}^{\text{bh}} \leq B_u^{\text{bh}} g_{k,u,\tau}, & \forall (k, \tau) \in \mathcal{I}, \forall u \in U, \forall t \geq \tau, \\ h_{k,u,\tau} = p_{k,u,\tau} + g_{k,u,\tau}. \end{cases} \quad (17)$$

Eqs. (16) and (17) impose mode consistency on computing and communication allocations by requiring each allocation to be zero unless the corresponding execution mode is selected.

(iii) *Multi-slot sufficiency for completion* Define $\chi_{k,\tau}(t) \triangleq \sum_{r=\tau}^t \delta_{k,\tau,r} \in \{0, 1\}$, which indicates whether task (k, τ) has been completed by the end of slot t . Then, for any $t \geq \tau$, the cumulative service must be sufficient whenever

completion by slot t is claimed. Then, we have following constraints:

$$\begin{cases} \ell_{k,\tau} = 1 \Rightarrow \sum_{s=\tau}^t f_{k,\tau,s}^{\text{loc}} \Delta \geq W_{k,\tau} \chi_{k,\tau}(t), \\ p_{k,u,\tau} = 1 \Rightarrow \sum_{s=\tau}^t f_{k,u,\tau,s}^{\text{uav}} \Delta \geq W_{k,\tau} \chi_{k,\tau}(t), \\ g_{k,u,\tau} = 1 \Rightarrow \sum_{s=\tau}^t f_{k,u,\tau,s}^{\text{cld}} \Delta \geq W_{k,\tau} \chi_{k,\tau}(t), \\ \text{where, } \forall (k,\tau) \in \mathcal{I}, \quad \forall t \geq \tau, \quad \forall u \in U. \end{cases} \quad (18)$$

$$\begin{cases} h_{k,u,\tau} = 1 \Rightarrow \sum_{s=\tau}^t \gamma_{k,u,s}^{\text{ul}} b_{k,u,\tau,s}^{\text{ul}} \Delta \geq B_{k,\tau} \chi_{k,\tau}(t), \\ g_{k,u,\tau} = 1 \Rightarrow \sum_{s=\tau}^t \gamma_{k,u,s}^{\text{bh}} b_{k,u,\tau,s}^{\text{bh}} \Delta \geq B_{k,\tau} \chi_{k,\tau}(t), \\ \text{where, } \forall (k,\tau) \in \mathcal{I}, \quad \forall t \geq \tau, \quad \forall u \in U. \end{cases} \quad (19)$$

Eq. (18) ensures that a task can be declared completed only after receiving enough cumulative computing service. Eq. (19) imposes the same idea on communication: the cumulative transmitted bits must be no smaller than the task input size whenever completion is claimed.

d) *Service-window indicators:*

$$\sum_{m \in S} \eta_{u,m,t} \leq 1, \quad \forall u \in U, \quad \forall t \in \mathcal{T}, \quad (20)$$

$$\eta_{u,m,t} \leq y_{u,m}, \quad \forall u \in U, \quad \forall m \in S, \quad \forall t \in \mathcal{T}, \quad (21)$$

$$\eta_{u,m,t} = 1 \Rightarrow a_{u,m} \leq t\Delta \leq d_{u,m}, \quad (22)$$

where, $\forall u \in U, \quad \forall m \in S, \quad \forall t \in \mathcal{T}$.

Eqs. (20) to (22) define the slot-level service-window indicators by allowing service only at assigned stations and within the corresponding arrival-departure interval.

e) *Service-window feasibility for UAV-involved processing:* Let $m(k)$ denote the station to which ISD k belongs. Then,

$$\begin{aligned} b_{k,u,\tau,t}^{\text{ul}} &\leq B_u^{\text{ul}} \eta_{u,m(k),t}, & f_{k,u,\tau,t}^{\text{uav}} &\leq F_u^{\text{uav}} \eta_{u,m(k),t}, \\ b_{k,u,\tau,t}^{\text{bh}} &\leq B_u^{\text{bh}} \eta_{u,m(k),t}, & \forall (k,\tau) \in \mathcal{I}, \quad \forall u \in U, \quad \forall t \geq \tau. \end{aligned} \quad (23)$$

Eq. (23) requires the UAV-involved communication and computing allocations of task (k,τ) in slot t to be zero unless UAV u is serving station $m(k)$ in that slot.

f) *Deadline constraint:*

$$z_{k,\tau} = 1 \Rightarrow T_{k,\tau} \leq \tau\Delta + D_k, \quad \forall (k,\tau) \in \mathcal{I}. \quad (24)$$

Eq. (24) states that a task is marked as on-time only if its completion time is no later than its deadline.

g) *UAV energy budget:* The total energy of UAV u is decomposed into flight/hover energy, onboard computing

energy, and communication energy:

$$E_u^{\text{fly}} = \alpha_{\text{fly}} \sum_{i \in L} \sum_{j \in L} d_{i,j} x_{u,i,j} + \alpha_{\text{hov}} \sum_{m \in S} (d_{u,m} - a_{u,m}) y_{u,m}, \quad (25)$$

$$E_u^{\text{cmp}} = \alpha_{\text{cmp}} \sum_{t \in \mathcal{T}} \sum_{(k,\tau) \in \mathcal{I}: \tau \leq t} f_{k,u,\tau,t}^{\text{uav}} \Delta, \quad (26)$$

$$E_u^{\text{com}} = \alpha_{\text{ul}} \sum_{t \in \mathcal{T}} \sum_{(k,\tau) \in \mathcal{I}: \tau \leq t} \gamma_{k,u,t}^{\text{ul}} b_{k,u,\tau,t}^{\text{ul}} \Delta + \alpha_{\text{bh}} \sum_{t \in \mathcal{T}} \sum_{(k,\tau) \in \mathcal{I}: \tau \leq t} \gamma_{k,u,t}^{\text{bh}} b_{k,u,\tau,t}^{\text{bh}} \Delta. \quad (27)$$

The total energy must satisfy

$$E_u^{\text{fly}} + E_u^{\text{cmp}} + E_u^{\text{com}} \leq E_u^{\text{max}}, \quad \forall u \in U. \quad (28)$$

Eqs. (25) to (27) define the flight/hover, onboard computing, and communication energy components of UAV u , and Eq. (28) imposes the corresponding per-UAV energy budget over the whole mission.